

Machine Learning Analysis of EEG-Based Seizure Prediction

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1 Introduction

Epileptic seizure prediction using EEG data is a challenging problem in medical machine learning. Model performance depends heavily on preprocessing, feature representation, regularisation, and class imbalance. This project investigates how these factors affect logistic regression performance across multiple EEG datasets.

2 Datasets

- Epileptic Seizure Recognition (highly imbalanced)
- EEG Signal Dataset (balanced)
- ADHD EEG Dataset (moderate complexity)

3 Methodology

Two preprocessing pipelines were used:

- Pipeline A: Feature selection
- Pipeline B: PCA transformation

Regularisation techniques:

- L1
- L2
- Elastic Net

Imbalance handling:

- Class weighting
- SMOTE
- Undersampling

4 Results

4.1 Baseline Performance

Dataset	Pipeline	Accuracy	Precision	Recall	F1-score
Epileptic	A	0.8094	1.0000	0.0470	0.0897
Epileptic	B	0.8021	1.0000	0.0104	0.0207
EEG	A	0.9600	0.9167	0.8800	0.8980
EEG	B	0.9600	0.9167	0.8800	0.8980
ADHD	A	0.6452	0.6923	0.5625	0.6207
ADHD	B	0.6452	0.6471	0.6875	0.6667

The EEG dataset achieved the best performance, while the epileptic dataset showed extremely low recall due to severe class imbalance.

4.2 Regularisation Analysis

Dataset	Regularisation	F1-score	PR-AUC
Epileptic	L1	0.0897	0.4315
Epileptic	L2	0.0897	0.4358
Epileptic	Elastic Net	0.0896	0.4337
EEG	L1	0.8980	0.9699
EEG	L2	0.8980	0.9739
EEG	Elastic Net	0.8980	0.9733
ADHD	L1	0.6667	0.7198
ADHD	L2	0.6207	0.7452
ADHD	Elastic Net	0.6875	0.7714

L2 regularisation provided the most stable performance across datasets, while Elastic Net performed best for the ADHD dataset.

4.3 Effect of Class Imbalance

Method	Precision	Recall	F1-score
None	1.0000	0.0470	0.0897
Class Weight	0.2681	0.4713	0.3417
SMOTE	0.2590	0.4748	0.3352
Undersampling	0.2579	0.4852	0.3368

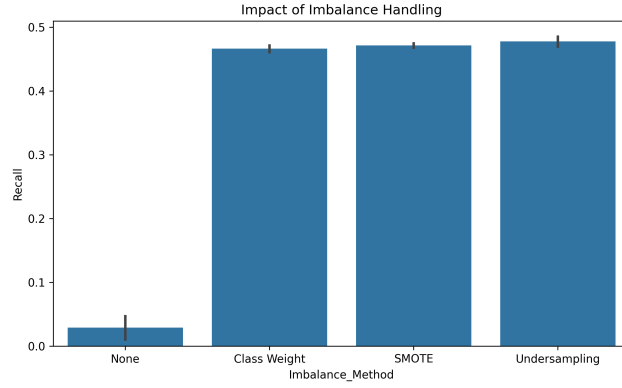


Figure 1: Impact of imbalance handling on recall (Epileptic dataset)

Without imbalance handling, recall was extremely low. After applying balancing techniques, recall improved significantly to around 0.47–0.48.

4.4 Precision-Recall Analysis

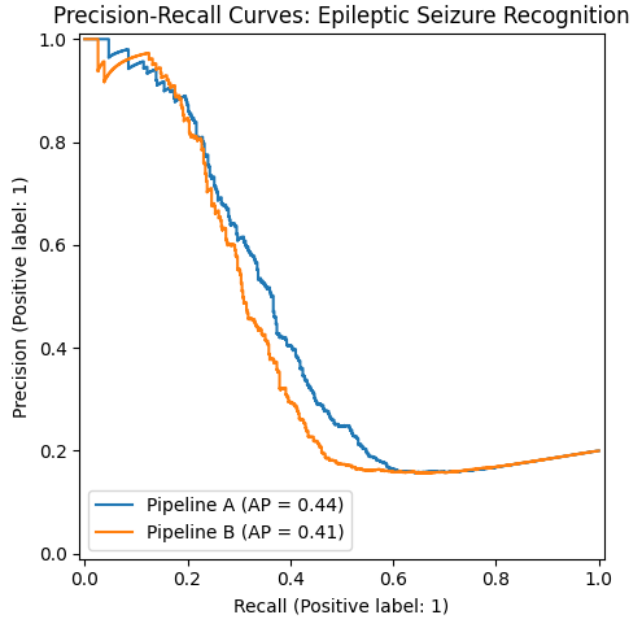


Figure 2: Precision-Recall curve (Epileptic dataset)

Precision decreases sharply as recall increases, showing difficulty in detecting minority seizure cases.

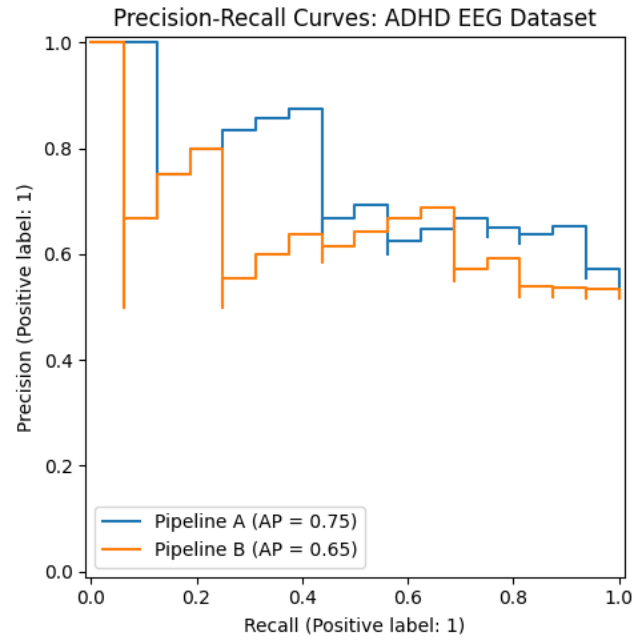


Figure 3: Precision-Recall curve (ADHD dataset)

Pipeline A performs better than Pipeline B for the ADHD dataset.

4.5 Learning Curve Analysis

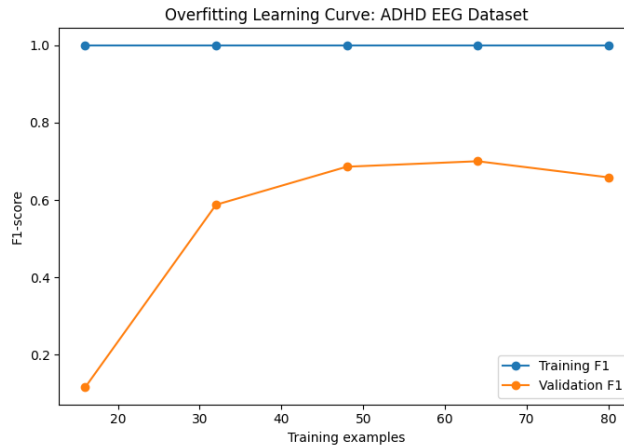


Figure 4: Overfitting behaviour (ADHD dataset)

The ADHD dataset shows clear overfitting: training performance is very high while validation performance is lower.

5 Discussion

Pipeline A generally outperformed Pipeline B, indicating that feature selection is more effective than PCA for EEG data.

L2 regularisation provided stable performance across datasets, while Elastic Net improved performance in the ADHD dataset.

Class imbalance had the largest impact. Without correction, the model failed to detect seizure cases. After applying imbalance handling, recall improved significantly but precision decreased, showing a trade-off.

6 Conclusion

This study demonstrates that class imbalance is the most important factor affecting model performance. Proper imbalance handling significantly improves recall. Pipeline A and L2 regularisation provide the most consistent performance across datasets.